

Concurrent Programming - Exercícios 3

CCS

1. Which of the following expressions of CCS are correct?
 - (a) $(a:0 + \bar{a}:A)nfa;bg$
 - (b) $(a:0 + \bar{a}:A)nfa;\tau g$
 - (c) $\tau:\tau:B + 0$
 - (d) $(a:b:B + \bar{a}:0)jB$
 - (e) $(a:b:B + \bar{a}:0):B$
2. Let $A := b:a:B$, using the inference rules show that the following transitions exist:
 - $(A\bar{b}:0)nfbg \xrightarrow{\tau} (a:Bj0)nfbg$
3. Solve the CCS exercises in PseuCo.com
4. Consider the following definitions of a researcher that takes coffee and publishes articles.

$$\begin{aligned}
 CM &:= coin?:coffee!:CM \\
 CS &:= pub!:coin!:coffee?:CS \\
 Uni &:= (CMjCS)nfcoint;coffeeeg
 \end{aligned}$$

Use the CSS inference rules to obtain the reachable fragment of $\llbracket Uni \rrbracket_{\Gamma}$. Test in pseuco.com. Compare with $Spec := pub!:Spec$.

5. Let $A := (a:A)nfbg$ show that $\llbracket A \rrbracket_{\Gamma}$ infinite (even the reachable fragment).
6. The following definitions try to solve the mutual exclusion with a semaphore.
 - (a)

$$\begin{aligned}
 Mutex_1 &:= (UserjSem)nfp;vg \\
 User &:= p:enter:exit:v:User \\
 Sem &:= p:v:Sem
 \end{aligned}$$

Use the CCS inference rules to obtain the reachable fragment of $\llbracket Mutex_1 \rrbracket_{\Gamma}$. Test in pseuco.com.

- (b) Let

$$Mutex_2 := ((UserjSem)jUser)nfp;vg$$

Use the CCS inference rules to obtain the reachable fragment of $\llbracket Mutex_2 \rrbracket_{\Gamma}$. Test in pseuco.com. There will be changes if one uses $User := p:enter:\bar{v}:exit:User$?

